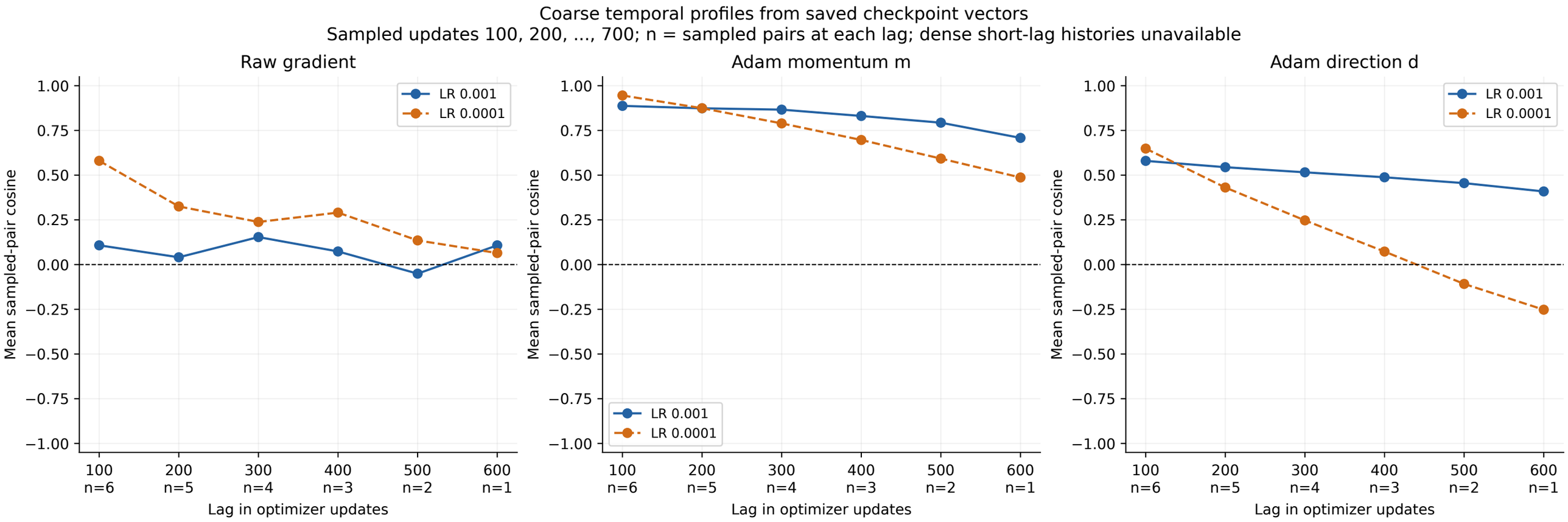

Coarse temporal profiles from saved checkpoint vectors
Sampled updates 100, 200, ..., 700; n = sampled pairs at each lag; dense short-lag histories unavailable
Raw gradient
Adam momentum m
Adam direction d
Mean sampled-pair cosine
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
LR 0.001
LR 0.0001
100
n=6
200
n=5
300
n=4
400
n=3
500
n=2
600
n=1
Lag in optimizer updates